

Coupling Polytopes and Toric Geometry: the 2×3 Family, the Square Case, Higher Levels and Stability

A companion to *Conditionals, Couplings and Toric Geometry*

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Abstract

A coupling of two finite distributions is a point of a transportation polytope. For nondegenerate marginals this polytope is Delzant [2], so it is the moment polytope of a compact toric symplectic manifold. This companion works the correspondence out in the cases that can be computed completely, and records how far it goes.

- *The 2×3 family.* The six walls $f_i = g_j$ cut the space of marginals into 18 chambers, which fall into 5 orbits under the symmetry group $S_2 \times S_3$. The surfaces that occur are $\mathbb{CP}^2$, $\mathbb{CP}^1 \times \mathbb{CP}^1$, the Hirzebruch surface F_1 , $\mathbb{CP}^2$ blown up at two points, and the del Pezzo surface of degree 6. Every facet normal lies in the fan of the last, so every surface in the family is the del Pezzo surface or a toric blow-down of it. Crossing a single wall is exactly one toric blow-up or blow-down.
- *The Kähler class.* It is the product coupling read as divisor coefficients, $[\omega_{f,g}] = \sum_c f_i g_j [D_c]$. Edge lengths of the coupling polygon are degrees of ω on toric curves, and its area is the symplectic volume.
- *The square 2×2 case.* With the unital whiskering, explicit whiskering defects are nonzero, and every defect is a tangent vector of a coupling polytope.
- *Higher levels.* The Delzant theorem applies at levels two and three. Flattening a level-two coupling is affine and commutes with stochastic change of base, but it is neither surjective nor injective in general.
- *Disc potentials.* They are sums over facets, which are cells, not cycles. For the degree-6 del Pezzo surface the potential has six critical points, all over the product coupling, with critical values $6, -2, -2, -2, -3, -3$. Square nondegenerate polytopes with $n \geq 3$ give manifolds that are not Fano. Optimal transport minimizes a moment-map component, so its optima are torus-fixed points, that is, spanning-tree couplings.
- *Stability.* Each $\omega_{f,g}$ gives a Bridgeland stability condition on the derived category of the corresponding surface, with explicit central charges. We state what this does and does not connect.
- *Cumulants.* Centred couplings are second mixed cumulants, and level-two flattening keeps only the cross-covariance of random measures. The whiskering defects are the Boolean cumulants

*Verification script: `check.polytope.companion.py` (26 checks; exact rational arithmetic except where marked). This version replaces the draft “Coupling polytopes and the 2×3 family”; [appendix A](#) lists what was corrected and what was withdrawn.

of whiskering, viewed as a noncommutative expectation. They are circulations, so tangent vectors of coupling polytopes. A cone in the sense of homotopy probability theory makes whiskering an A_∞ -functor through arity three, and its cohomology keeps only the marginals. Into the square-zero cone, whiskering is an A_∞ -functor with exactly two components.

- *The coupling nerve.* Couplings form the simplicial object $D(Y^{\bullet+1})$. Its 2-horns always fill, and the fillers form products of coupling polytopes, with the gluing filler the maximum-entropy one. Its 3-horns and 4-horns can fail to fill, with contextual fraction $\frac{1}{2}$ and $\frac{3}{4}$, so it is not Kan. Boundary fibres are Delzant for shape $2 \times n \times m$, orbifold polytopes for $3 \times 3 \times 3$, and eventually arbitrary.

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1 Introduction

The companion paper [2] proves that the polytope $\text{Cpl}(f, g)$ of couplings of two nondegenerate marginals is Delzant. By Delzant’s theorem [9] there is therefore a compact toric symplectic manifold $X_{f, g}$ whose moment map has image $\text{Cpl}(f, g)$. Each interior coupling is the image of a Lagrangian torus, and each vertex is the image of a fixed point. This note asks what the correspondence gives in concrete cases.

Section 2 fixes conventions and proves a general formula for the Kähler class. Section 3 treats the 2×3 family completely. Section 4 treats the square 2×2 case, the base case for the whiskering questions of [2, §7]. Section 5 extends the polytope results to second- and third-order states. Section 6 computes disc potentials and relates optimal transport to the moment map. Section 7

constructs stability conditions from marginals and states the remaining questions. [Section 8](#) records the status of the A_∞ question, [section 9](#) connects the polytopes to cumulants and homotopy probability theory, and [section 10](#) computes the low-dimensional fillers of the coupling nerve. Every numbered claim is checked in `check.polytope.companion.py`.

2 Setup

Let $f \in D(Y)$ and $g \in D(Y')$ have full support, with $|Y| = n$ and $|Y'| = m$, and let

$$\text{Cpl}(f, g) = \left\{ \alpha \in \mathbb{R}_{\geq 0}^{Y \times Y'} : \sum_{y'} \alpha(y, y') = f(y), \sum_y \alpha(y, y') = g(y') \right\}.$$

The tangent space is the space of circulations, that is matrices with zero row and column sums. The integer circulations form a lattice Λ of rank $(n-1)(m-1)$ with basis $e_{ij} = E_{ij} - E_{im} - E_{nj} + E_{nm}$ ($i < n, j < m$). Coordinates in this basis are the entries in the top-left $(n-1) \times (m-1)$ block. The pair (f, g) is *degenerate* if $\sum_I f = \sum_J g$ for some nonempty proper $I \subsetneq Y$ and $J \subsetneq Y'$. These hyperplanes are the *walls*, and the components of their complement are the *chambers*.

Theorem 2.1 ([2, Thm. 4.4]). *If (f, g) is nondegenerate, every vertex of $\text{Cpl}(f, g)$ is supported on a spanning tree of $K_{n,m}$, the edges at a vertex are the fundamental cycles of that tree, and $\text{Cpl}(f, g)$ is Delzant with respect to Λ .*

Facets and normals. The facets are among the sets $\{\alpha_c = 0\}$, one for each cell $c = (i, j)$. A cell gives no facet when $f_i + g_j > 1$, since then $\alpha_c \geq f_i + g_j - 1 > 0$. The inward normal of $\{\alpha_c = 0\}$ is the coordinate functional $\alpha \mapsto \alpha_c$ restricted to Λ . In the dual basis $\{e_{ij}^*\}$ it is

$$v_{(i,j)} = e_{ij}^*, \quad v_{(i,m)} = -\sum_j e_{ij}^*, \quad v_{(n,j)} = -\sum_i e_{ij}^*, \quad v_{(n,m)} = \sum_{i,j} e_{ij}^* \quad (i < n, j < m).$$

Each v_c is primitive, because some integer circulation has entry 1 at c .

Proposition 2.2 (The Kähler class is the product coupling). *For nondegenerate (f, g) , let D_c be the toric divisor of the facet $\{\alpha_c = 0\}$. Up to the normalization of ω , the symplectic class of $X_{f,g}$ is*

$$[\omega_{f,g}] = \sum_{c \text{ facet}} f_i g_j [D_c], \quad c = (i, j),$$

and $\omega_{f,g} \cdot C$ is the lattice length of the edge of $\text{Cpl}(f, g)$ that corresponds to a torus-invariant curve C .

Proof. For a Delzant polytope $\{u : \langle u, v_c \rangle + \lambda_c \geq 0\}$ the symplectic class is $\sum_c \lambda_c [D_c]$ [13]. Here the facet functions are $\alpha_c(u) = \langle u - u_0, v_c \rangle + \alpha_c(u_0)$ for any interior u_0 , so $\lambda_c = \alpha_c(u_0)$. Taking $u_0 = f \otimes g$ gives the formula. Another base point changes the coefficients by $\langle w, v_c \rangle$, which is a linear relation among the $[D_c]$. The degree of ω on a torus-invariant curve is the lattice length of its edge. $\square$

3 The 2×3 family

Write $f = (f_1, 1 - f_1)$, $g = (g_1, g_2, 1 - g_1 - g_2)$, $x = \alpha_{11}$ and $y = \alpha_{12}$. The polygon $\text{Cpl}(f, g)$ is cut out by

$$x \geq 0, \quad y \geq 0, \quad x \leq g_1, \quad y \leq g_2, \quad x + y \leq f_1, \quad x + y \geq f_1 + g_1 + g_2 - 1,$$

with inward normals $e_1, e_2, -e_1, -e_2, -(e_1 + e_2)$ and $e_1 + e_2$. The only proper subsets of a two-point and a three-point set that can have equal sums give the six walls $f_i = g_j$.

Theorem 3.1 (Chambers and surfaces). *The six walls cut the prism $\{0 < f_1 < 1, g_1, g_2 > 0, g_1 + g_2 < 1\}$ into 18 chambers. On each chamber the polygon type is constant and Delzant, and the surfaces are:*

<i>Polygon</i>	<i>Chambers</i>	<i>Self-intersections D_c^2</i>	<i>Surface</i>	<i>Orbit under $S_2 \times S_3$</i>
<i>triangle</i>	<i>2</i>	<i>(1, 1, 1)</i>	$\mathbb{CP}^2$	<i>one orbit of size 2</i>
<i>quadrilateral</i>	<i>3</i>	<i>(0, 0, 0, 0)</i>	$\mathbb{CP}^1 \times \mathbb{CP}^1$	<i>one orbit of size 3</i>
<i>quadrilateral</i>	<i>6</i>	<i>(1, 0, -1, 0)</i>	$F_1 = \text{Bl}_1 \mathbb{CP}^2$	<i>one orbit of size 6</i>
<i>pentagon</i>	<i>6</i>	<i>(0, -1, -1, -1, 0)</i>	$\text{Bl}_2 \mathbb{CP}^2$	<i>one orbit of size 6</i>
<i>hexagon</i>	<i>1</i>	<i>$(-1)^6$</i>	<i>del Pezzo of degree 6</i>	<i>fixed</i>

Here S_2 swaps the two rows and S_3 permutes the columns.

Proof. For an arrangement of hyperplanes restricted to a convex region, each realizable sign vector is exactly one convex chamber. A rational grid and random sampling realize the same 18 sign vectors. In each chamber the vertices are computed exactly, and the number of vertices and the self-intersections are constant on all samples. Self-intersections come from the relations $v_{i-1} + v_{i+1} = -D_i^2 v_i$ between cyclically ordered normals. Consistency is confirmed by $\sum_i D_i^2 = 12 - 3k$ for k divisors. Delzant means that consecutive normals have determinant ± 1 . The orbits are computed by letting $S_2 \times S_3$ act on sign vectors. $\square$

Theorem 3.2 (Every surface is a blow-down of the del Pezzo surface). *For every nondegenerate pair in the 2×3 family, $X_{f,g}$ is the del Pezzo surface of degree 6 or is obtained from it by toric blow-downs.*

Proof. The fan Σ of $X_{f,g}$ is smooth and complete. Its rays are among the six rays $\pm e_1, \pm e_2, \pm(e_1 + e_2)$ of the del Pezzo fan Σ_6 . In cyclic order each ray of Σ_6 is the sum of its neighbours. Two of the six rays form a lattice basis exactly when they are at cyclic distance one or two; rays at distance three are opposite. So adjacent rays u, w of Σ are at distance one or two, and in the second case the omitted ray between them is $u + w$. Inserting $u + w$ is the star subdivision that blows up the corresponding fixed point [11]. Inserting all omitted rays turns Σ into Σ_6 . $\square$

Proposition 3.3 (Wall-crossing). *On the wall $f_i = g_j$ three facet lines meet at a point. For example, on $f_1 = g_1$ the lines $x = g_1, y = 0$ and $x + y = f_1$ meet at $(g_1, 0)$. Crossing one wall cuts or restores one corner, which is one toric blow-up or blow-down. There are no flips, which do not occur for surfaces, and the type does not change inside a chamber.*

Proof. The concurrency is direct. For all 30 pairs of chambers whose sign vectors differ in one wall, the vertex counts differ by exactly one. Constancy inside a chamber is [2, Prop. 4.6]. $\square$

Example 3.4.

- (a) At $(f_1, g_1, g_2) = (0.1, 0.5, 0.2)$ the polygon is the triangle with vertices $(0, 0)$, $(0.1, 0)$ and $(0, 0.1)$, and the surface is $\mathbb{CP}^2$.
- (b) At $(0.4, 0.5, 0.3)$ the polygon is the pentagon with vertices $(0, 0.2)$, $(0, 0.3)$, $(0.1, 0.3)$, $(0.4, 0)$ and $(0.2, 0)$, and the surface is $\text{Bl}_2 \mathbb{CP}^2$. The straight path to the uniform point crosses only the wall $f_1 = g_1$; the corner at $(f_1, 0)$ is cut and the hexagon appears.
- (c) At the uniform point $(\frac{1}{2}, \frac{1}{3}, \frac{1}{3})$ the hexagon has vertices $(0, \frac{1}{6})$, $(0, \frac{1}{3})$, $(\frac{1}{6}, \frac{1}{3})$, $(\frac{1}{3}, \frac{1}{6})$, $(\frac{1}{3}, 0)$ and $(\frac{1}{6}, 0)$, all six edges have lattice length $\frac{1}{6}$, its lattice area is $\frac{1}{12}$, and $[\omega] = -\frac{1}{6}K_X$. This is the unique monotone point of the family [2, Prop. 4.7]: monotonicity holds at a point, not on a chamber.

Proposition 3.5 (Kähler data in the family). *In all 18 chambers, $\omega_{f,g} \cdot D_c$ equals the lattice length of the edge $\{\alpha_c = 0\}$ and is positive, so $\omega_{f,g}$ is ample, and $\frac{1}{2}\omega_{f,g}^2$ equals the lattice area of $\text{Cpl}(f, g)$.*

Proof. These are [proposition 2.2](#) and the Duistermaat–Heckman formula for the volume. The script verifies both in exact arithmetic, using the intersection matrix of each fan. $\square$

4 The square 2×2 case

Let $B = B' = \{0, 1\}$. A 1-cell is a stochastic matrix $k(a, b)$ with rows $(1 - a, a)$ and $(1 - b, b)$, and Kleisli composition is matrix multiplication. The deterministic 1-cells are the constants $e_0 = k(0, 0)$ and $e_1 = k(1, 1)$, the identity $\sigma = k(0, 1)$ and the swap $\tau = k(1, 0)$. A coupling 2-cell $\alpha: f \Rightarrow g$ is a coupling $\alpha(x)$ of $f(x)$ and $g(x)$ at each x . Its cost is $c(\alpha) = \max_x \Pr_{\alpha(x)}(y \neq y')$. Vertical composition is the gluing lemma, strictly associative with diagonal identities [2, Thm. 2.3].

Proposition 4.1 (Coupling segments and costs). *For $f(x) = (p, 1 - p)$ and $g(x) = (q, 1 - q)$, the couplings are $\begin{pmatrix} a & p-a \\ q-a & 1-p-q+a \end{pmatrix}$ with $\max(0, p+q-1) \leq a \leq \min(p, q)$. The product coupling has $a = pq$, and the maximal coupling, of cost $\text{TV}(f(x), g(x))$, has $a = \min(p, q)$. Between deterministic 1-cells the coupling is unique, and its cost is 0 for equal 1-cells and 1 for any two distinct ones, because distinct deterministic maps differ at some x and the cost is a maximum over x .*

Definition 4.2 (Whiskering). Precomposition by a deterministic $u: W \rightarrow B$ is $(\alpha \circ u)(w) = \alpha(u(w))$. Postcomposition by $h: B' \rightarrow D(B'')$ copies h on equal pairs and runs it independently on unequal pairs:

$$(h \cdot \alpha)(x)(z, z') = \sum_y \alpha(x)(y, y) h(y)(z) [z = z'] + \sum_{y \neq y'} \alpha(x)(y, y') h(y)(z) h(y')(z').$$

This is the whiskering of [2, Thm. 2.3(ii)]. It sends diagonal couplings to diagonal couplings. Running h independently also on equal pairs would not preserve identities: for $h = k(\frac{9}{10}, \frac{1}{10})$ it sends $\frac{1}{2}\Delta$ to $(\frac{0.41}{0.09}, \frac{0.09}{0.41})$, so it cannot be the first component of any functor.

Proposition 4.3 (Whiskering defects). *Let $F_2(\beta, \alpha) = (h \cdot \beta) \cdot (h \cdot \alpha) - h \cdot (\beta \cdot \alpha)$.*

- (i) *Precomposition by a deterministic map is strictly functorial.*
- (ii) *If α is an identity, then $F_2(\beta, \alpha) = 0$.*
- (iii) *With $f = g$ having uniform rows and $\alpha = \beta$ the swap coupling at each x , $F_2(\beta, \alpha)(x) = t \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$, where $t = \frac{369}{2500}$ for $h = k(\frac{9}{10}, \frac{1}{10})$ and $t = \frac{1}{4}$ for h constant and uniform.*
- (iv) *Every defect has zero row and column sums. It is a tangent vector of the coupling polytope of $h^\# f(x)$ and $h^\# k(x)$, not a coupling.*

Proof. (i) Both sides evaluate the same couplings at $u(w)$. (ii) $h \cdot \Delta = \Delta$, and Δ is a unit for gluing. (iii) $\beta \cdot \alpha = \text{swap} \cdot \text{swap} = \Delta$, so $h \cdot (\beta \cdot \alpha)$ is diagonal, while $h \cdot \text{swap}$ has mass off the diagonal; the values are exact computations. (iv) Both terms couple the same two marginals. $\square$

So the defect lives in the tangent space of a coupling polytope. There the Fisher metric, which is twice the Hessian of Guillemin’s potential [2, Prop. 4.5], gives it a size. This is the only sense in which the defect meets the toric geometry that we can establish.

5 Second- and third-order states

Let $Y \subseteq DB$ be a finite set of n first-order states and $P, Q \in D(Y)$ second-order states with full support. A level-two coupling is $\Pi \in D(Y \times Y)$ with marginals P and Q , and $\text{Cpl}_2(P, Q)$ is the transportation polytope on $K_{n,n}$ with these marginals.

Proposition 5.1 (Delzant at levels two and three). *If (P, Q) is nondegenerate, then $\text{Cpl}_2(P, Q)$ is Delzant of dimension $(n - 1)^2$. The same holds for third-order states $R, S \in D(Z)$, with Z a finite set of second-order states.*

Proof. [Theorem 2.1](#) uses only the marginals, not what the points of Y or Z are. $\square$

Flattening. The map

$$\mu^*: \text{Cpl}_2(P, Q) \rightarrow \text{Cpl}(\mu_B P, \mu_B Q), \quad \mu^*(\Pi)(b, b') = \sum_{\rho, \rho'} \Pi(\rho, \rho') \rho(b) \rho'(b'),$$

is affine. Its image contains $\mu_B P \otimes \mu_B Q$, the image of $P \otimes Q$.

Proposition 5.2 (Properties of flattening).

- (i) *If Y consists of point masses, μ^* is the identity.*
- (ii) *μ^* is not surjective in general: for $Y = \{\rho\}$ with ρ uniform on two points the image is $\{\rho \otimes \rho\}$, while $\text{Cpl}(\rho, \rho)$ is a segment containing the diagonal coupling.*
- (iii) *For $n = 2$ and $\rho_1 \neq \rho_2$, μ^* is injective: the segment direction maps to $(\rho_1 - \rho_2) \otimes (\rho_1 - \rho_2) \neq 0$.*
- (iv) *If $(n - 1)^2 > (|B| - 1)^2$, the fibres of μ^* are positive-dimensional. For example, with three states on two outcomes and uniform $P = Q$ the fibre has dimension 3, so level-two couplings carry information that level-one couplings lose.*
- (v) *For a kernel $k: B \rightarrow D(B')$, pushing Π forward along $k^\# \times k^\#$ commutes with flattening: $\mu^*((k^\# \times k^\#)_* \Pi) = (k \otimes k)^\# \mu^*(\Pi)$. This is the level-two form of stochastic change of base [4, Prop. 9.14].*

Proof. (i) $\rho(b) = [\rho = \delta_b]$. (ii) and (iii) are direct. (iv) The image lies in a space of dimension $(|B| - 1)^2$; the example is computed exactly. (v) Both sides equal $\sum \Pi(\rho, \rho') k^\# \rho \otimes k^\# \rho'$. $\square$

Uniform marginals. Uniform $n \times n$ marginals are degenerate for every $n \geq 2$. For $n = 2$ the polytope is still a segment, so $X = \mathbb{CP}^1$ and it is monotone, with special point the product coupling. For $n \geq 3$ the polytope is not simple and has no smooth toric manifold.

6 Disc potentials and optimal transport

For a Delzant polytope with facet functions ℓ_c and normals v_c , the potential of the torus fibre over u is $W_u(z) = \sum_c T^{\ell_c(u)} z^{v_c}$ over facets, for compact toric Fano manifolds [7] and, with corrections, for general compact toric manifolds [10]. A fibre with a flat connection has nonvanishing Floer cohomology when it corresponds to a critical point. For coupling polytopes the facets are cells, so W has at most nm terms, with exponents given by the normals of [section 2](#). The facets are not the cycles of $K_{n,m}$.

Proposition 6.1 (The del Pezzo potential). *At the uniform 2×3 point all facet functions equal $\frac{1}{6}$ at the product coupling, and $W = T^{1/6}(x + y + x^{-1} + y^{-1} + xy + x^{-1}y^{-1})$. Up to the factor $T^{1/6}$, W has six critical points: $(1, 1)$, $(1, -1)$, $(-1, 1)$, $(-1, -1)$, (ζ, ζ) and (ζ^2, ζ^2) for a primitive cube root of unity ζ . All six have $|x| = |y| = 1$, so they lie over the product coupling and differ only in the flat connection. The critical values are $6, -2, -2, -2, -3, -3$.*

Proof. Exact solution of $x\partial_x W = y\partial_y W = 0$; for instance $W(\zeta, \zeta) = 3\zeta + 3\zeta^2 = -3$. The number 6 equals twice the area 3 of the Newton hexagon, as Kushnirenko’s theorem predicts. $\square$

Example 6.2 (Non-monotone cases).

- (a) *A segment.* For the 2×2 segment $[\max(0, p + q - 1), \min(p, q)]$, $X = \mathbb{CP}^1$ and the special fibre is the midpoint. That is the product coupling $a = pq$ when $p = \frac{1}{2}$, but not in general: for $p = q = \frac{1}{3}$ the midpoint is $\frac{1}{6}$ and $pq = \frac{1}{9}$.
- (b) *A 3×3 case.* For $P = (0.40, 0.35, 0.25)$ and $Q = (0.30, 0.43, 0.27)$ the polytope is Delzant with 18 vertices and 9 facets, one per cell, so W has 9 terms.

Proposition 6.3 (Square cases are not Fano). *Let $n \geq 3$ and let (f, g) be a nondegenerate $n \times n$ pair for which every cell is a facet. Then $X_{f,g}$ is not Fano. In the 3×3 example above, $-K_X$ is nef and big, so X is weak Fano.*

Proof. $-K_X = \sum_c D_c$ corresponds to the polytope $\{\langle u, v_c \rangle \geq -1 \ \forall c\}$, which is the transportation polytope with all row and column sums n , that is n times the Birkhoff polytope. If $-K_X$ were ample, this polytope would have normal fan equal to the smooth fan of X and would be simple, but the Birkhoff polytope is not simple for $n \geq 3$. For nefness, each maximal cone (a spanning tree T) must give a vertex of that polytope, that is, the tree solution of T with all marginals equal to n must be nonnegative. This is checked for all 18 trees of the example. Bigness is full-dimensionality. $\square$

So Cho–Oh’s Fano theorem applies to the 2×3 family, whose surfaces are all Fano, but not to square cases with $n \geq 3$, where the general theory of [10] is needed.

Proposition 6.4 (Optimal transport is a moment-map problem). *For a cost $c: Y \times Y' \rightarrow \mathbb{R}$, the optimal-transport value $\min_{\alpha \in \text{Cpl}(f,g)} \langle c, \alpha \rangle$ is the minimum over $X_{f,g}$ of the function $\langle c, \mu \rangle$, where μ is the moment map. It is attained at a torus-fixed point, that is, a vertex: a coupling supported on a spanning tree. If c is integral on Λ , then $\langle c, \mu \rangle$ is the Hamiltonian of a circle action.*

Proof. μ maps $X_{f,g}$ onto $\text{Cpl}(f, g)$. A linear function on a polytope is minimized at a vertex, and the preimage of a vertex is a fixed point. A component $\langle \xi, \mu \rangle$ with ξ in the lattice generates the circle action of ξ [9]. The script checks the first statement by linear programming against vertex enumeration. $\square$

This is where the polytope picture meets the obstruction theory of [4]. Kantorovich comparisons between second-order states [4, Prop. 9.14], and distances to effective sets, are linear programs over polytopes of this kind. Their optima are spanning-tree couplings, and their dual variables are the certificates. These vertices are not “committed” states in the tower’s sense. A spanning-tree coupling has $n + m - 1$ nonzero entries and is not a point mass.

7 Stability conditions

A Bridgeland stability condition on a triangulated category is a pair $(Z, \mathcal{A})$: the heart $\mathcal{A}$ of a bounded t -structure, and an additive central charge $Z: K_0 \rightarrow \mathbb{C}$ that maps nonzero objects of $\mathcal{A}$ into the upper half-plane together with $\mathbb{R}_{<0}$, with Harder–Narasimhan filtrations and the support property [5]. The stability conditions form a complex manifold [5]. On any smooth projective surface, for any ample class ω , there is one with central charge $Z_\omega(E) = -\int e^{-i\omega} \text{ch}(E)$ and a tilted heart [1, 18]. (Bridgeland’s earlier construction [6] is for K3 surfaces.)

Proposition 7.1 (Marginals give stability conditions). *For (f, g) in a chamber of the 2×3 family, $\omega_{f,g}$ defines a stability condition $\sigma_{f,g}$ on $D^b(\text{Coh } X_{f,g})$, depending continuously on (f, g) within the chamber, with*

$$Z(\mathcal{O}_X) = \text{area Cpl}(f, g), \quad Z(\mathcal{O}_x) = -1, \quad Z(\mathcal{O}_{D_c}) = \frac{1}{2}D_c^2 + i \text{ length}\{\alpha_c = 0\}.$$

As (f, g) approaches a wall from inside a chamber, one edge length tends to 0, its divisor E is a (-1) -curve, and $Z(\mathcal{O}_E) \rightarrow -\frac{1}{2}$.

Proof. $\omega_{f,g}$ is ample by proposition 3.5. The values follow from $\text{ch}(\mathcal{O}_X) = 1$, $\text{ch}(\mathcal{O}_x) = [\text{pt}]$ and $\text{ch}(\mathcal{O}_D) = D - \frac{1}{2}D^2$, together with proposition 3.5. The statement about walls is proposition 3.3 with the self-intersections of theorem 3.1, and is checked in [2, §7.1]. $\square$

Mirror symmetry. For a toric Fano manifold the disc potential is the Hori–Vafa Landau–Ginzburg potential [7]. The disc potential enters central charges through oscillatory integrals $\int e^{-W/z}$ over integral cycles, whose asymptotics involve the $\widehat{\Gamma}$ -class [16, 12], not through its values. The critical values in proposition 6.1 are the eigenvalues of quantum multiplication by c_1 , which is what the Gamma conjectures concern.

Question 7.2.

- (i) Does the path $\sigma_{f,g}$ continue across a wall, glued along Orlov’s decomposition $D^b(\text{Bl}_p X) = \langle \mathcal{O}_E(-1), D^b(X) \rangle$ [19]?
- (ii) Is there a natural object attached to an interior coupling, for instance the mirror of its torus fibre with a flat connection, whose central charge has a probabilistic meaning?
- (iii) Do the quantum-corrected central charges with $\omega = \omega_{f,g}$ define stability conditions on these surfaces, and does the monotone point play a special role?

Not claimed. We make no identification of the disc potential with a central charge, of spanning-tree couplings with a heart, of the obstruction tower with Harder–Narasimhan filtrations, or of hallucination with instability.

8 The A_∞ question

The hom-categories of coupling 2-cells are strictly associative, so as A_∞ -categories they have $m_k = 0$ for $k \geq 3$. If a linearization also has $m_1 = 0$, the arity-two A_∞ -functor equation reduces to $F_1(m_2(\beta, \alpha)) = m_2(F_1\beta, F_1\alpha)$, which says that whiskering is functorial. Proposition 4.3(iii) shows that it is not. Hence the defect can be the second component of an A_∞ -functor only in a linearization with nonzero m_1 , for instance chains on coupling polytopes in which the defects

are boundaries [2, Conj. 7.3]. No such linearization is known. Statements identifying the defect with the Hessian of a disc potential, or the functor equation with a Picard–Lefschetz formula, are withdrawn (appendix A). Section 9 shows how a cone supplies such an m_1 .

9 Cumulants and the homotopy viewpoint

Homotopy probability theory treats an expectation that fails to be an algebra map as a morphism of homotopy algebras. Its components are then the cumulants [14, 15]. This section shows that the coupling polytopes carry exactly this structure. Couplings are second mixed cumulants, level-two flattening keeps only second cumulants, and the whiskering defects are the Boolean cumulants of the whiskering map. All of them are tangent vectors of coupling polytopes. Everything here is checked in `check_cumulants.py`.

Proposition 9.1 (Couplings are second cumulants). *For $\alpha \in \text{Cpl}(f, g)$, the centred coupling $\alpha - f \otimes g$ is the cross-covariance matrix $\kappa_2(1_{\{y=i\}}, 1_{\{y'=j\}})$ of the indicator vectors under α . So $\text{Cpl}(f, g) - f \otimes g$ is the set of cross-covariances compatible with the marginals and with positivity. It lies in the space of circulations, and the product coupling is the point where every mixed cumulant of the pair (y, y') vanishes. The product coupling is also the only point of $\text{Cpl}(f, g)$ of rank one, that is, the only point on the toric variety of the independence model. This is the Birch point of that model [20].*

Proof. $\mathbb{E}[1_{\{y=i\}}1_{\{y'=j\}}] - \mathbb{E}[1_{\{y=i\}}]\mathbb{E}[1_{\{y'=j\}}] = \alpha_{ij} - f_i g_j$. A nonnegative rank-one matrix $uv^\top$ with row sums f and column sums g equals $f \otimes g$. For two finite variables, independence is equivalent to vanishing of all mixed cumulants of the indicators. $\square$

Proposition 9.2 (Level two: flattening keeps the second cumulant). *For a level-two coupling $\Pi \in \text{Cpl}_2(P, Q)$,*

$$\mu^*(\Pi) = \mu_B P \otimes \mu_B Q + \text{Cov}_\Pi(\rho, \rho'),$$

where $\text{Cov}_\Pi(\rho, \rho') = \mathbb{E}_\Pi[(\rho - \mu_B P) \otimes (\rho' - \mu_B Q)]$ is the cross-covariance of the two random measures, a circulation. The fibres of μ^ are therefore the level-two couplings with a fixed cross-covariance. For three states on two outcomes, the mixed moments $\mathbb{E}_\Pi[\rho(0)^a \rho'(0)^b]$ with $a, b \leq 2$ determine Π , while flattening sees only $a = b = 1$; this accounts for the three-dimensional fibres of proposition 5.2(iv).*

Proof. Expand $\mathbb{E}_\Pi[\rho \otimes \rho']$. For the example, the map from the four-dimensional space of circulations to the mixed moments of order up to $(2, 2)$ has rank 4, and to the order- $(1, 1)$ moment it has rank 1. $\square$

So level-two couplings carry the higher mixed cumulants of a pair of random distributions, and level-one gluing keeps only the second. This is the cumulant form of “level-two gluing keeps correlations that level-one gluing loses” [3, Prop. 7.16].

Whiskering as an expectation. Fix $f \in D(Y)$ and let A_f be the space of real $Y \times Y$ matrices whose row sums and column sums are both multiples of f ; it is spanned by the couplings of f with itself. With the gluing product $a \cdot b = a \text{diag}(f)^{-1}b$ and unit $\text{diag}(f)$, A_f is an associative unital algebra. The circulations I_f form a two-sided ideal, and $A_f/I_f \cong \mathbb{R}$ records only the total mass. Whiskering by a kernel h (definition 4.2) is a linear unital map $h_*: A_f \rightarrow A_{h\#f}$, a noncommutative

“expectation” that need not be multiplicative. Its *Boolean cumulants* [21] are defined recursively over interval partitions:

$$\begin{aligned} b_1 &= h_*, & b_2(a, b) &= h_*(a \cdot b) - h_*(a) \cdot h_*(b), \\ b_3(a, b, c) &= h_*(a \cdot b \cdot c) - h_*a \cdot h_*b \cdot h_*c - b_2(a, b) \cdot h_*c - h_*a \cdot b_2(b, c). \end{aligned}$$

Here $a \cdot b$ means: glue a first, then b .

Proposition 9.3 (Whiskering defects are Boolean cumulants).

- (i) The whiskering defect of [proposition 4.3](#) is $F_2(\beta, \alpha) = -b_2(\alpha, \beta)$.
- (ii) Every b_n with $n \geq 2$ takes values in the circulations $I_{h\#f}$, that is, in tangent vectors of coupling polytopes. Modulo $I_{h\#f}$, h_* is an algebra map.
- (iii) For $|Y| = 3$ the defects b_2 span all of $I_{h\#f}$, and b_3 is nonzero in general.
- (iv) Let C be the differential graded algebra obtained from $A_{h\#f}$ by freely adjoining generators $\delta(a, b)$ of degree -1 with $d\delta(a, b) = b_2(a, b)$; this is a cone in the sense of homotopy probability theory. Then (h_*, δ) satisfies the arity-two equation of an A_∞ -functor $A_f \rightarrow C$, the arity-three obstruction is a cycle, and $H^0(C) = A_{h\#f}/(b_2)$, which equals $A_{h\#f}/I_{h\#f} \cong \mathbb{R}$ whenever the defects span $I_{h\#f}$, as in (iii).

Proof. (i) is the definition. (ii) Both composites couple the same two marginals, so their difference is a circulation; the recursion preserves this because I is an ideal. Modulo I only masses remain, and they multiply. (iii) The defects from 40 random instances have rank $4 = \dim I$ for three points, and random instances give $b_3 \neq 0$. (iv) The arity-two equation is $d\delta = b_2$. The arity-three obstruction is $h_*a \cdot \delta(b, c) - \delta(a, b) \cdot h_*c - \delta(a \cdot b, c) + \delta(a, b \cdot c)$, whose differential vanishes identically; this is verified exactly. H^0 of a free extension by degree -1 generators is the quotient by the two-sided ideal generated by their boundaries, here $I_{h\#f}$ by (iii). $\square$

Proposition 9.4 (A two-component A_∞ -functor into the square-zero cone). *Let $C = A_{h\#f} \oplus \varepsilon I_{h\#f}$, where ε has degree -1 , $\varepsilon^2 = 0$, $d(\varepsilon x) = x$, and $A_{h\#f}$ acts on εI through the ideal structure. Then C is a differential graded algebra with $H^0(C) = A_{h\#f}/I_{h\#f}$ and $H^{-1}(C) = 0$, so it is the mapping cone of $I \hookrightarrow A$, a model of the marginal quotient. Whiskering extends to an A_∞ -functor $A_f \rightarrow C$ with*

$$F_1 = h_*, \quad F_2(a, b) = \varepsilon b_2(a, b), \quad F_n = 0 \quad (n \geq 3).$$

For three and four points the defects span I , with rank $(n-1)^2$.

Proof. C is the trivial extension of A by the bimodule εI ; d is a derivation because I is an ideal, and d is injective on εI . In arity two the functor equation is $dF_2(a, b) = h_*(ab) - h_*a \cdot h_*b$, which holds by definition. In arity three, with $F_3 = 0$ (there is nothing in degree -2), it reads $b_2(ab, c) - b_2(a, bc) = h_*a \cdot b_2(b, c) - b_2(a, b) \cdot h_*c$. Expanding both sides gives $h_*a \cdot h_*(bc) - h_*(ab) \cdot h_*c$; this is the statement that b_2 is the Hochschild coboundary of h_* . In arity $n \geq 4$ every term contains F_k with $k \geq 3$ or a product $F_2 \cdot F_2 \in \varepsilon^2 I = 0$. The script verifies the arity-three identity exactly for $n = 2, 3, 4$, and the ranks for $n = 3, 4$. $\square$

So the answer to the first part of the question below is that no higher components are needed in this model. The higher Boolean cumulants b_n , which are nonzero, appear only in larger models such as the free cone. This parallels homotopy probability theory, where a cone factors the expectation as an inclusion followed by a quasi-isomorphism [14].

What this shows. Making whiskering an A_∞ -functor through a cone forgets exactly the correlations. The cohomology H^0 keeps only the marginals, that is, the Kleisli level of [2]. The correlation structure lives in the homotopy data: the Boolean cumulants b_n , which are tangent vectors of the coupling polytopes. In the geometry of section 2, the base of the fibration $\text{Cpl}(f, g) \mapsto (f, g)$ is the cohomological part, and the fibres, where the Fisher metric measures the cumulants [2, Prop. 4.5], carry the homotopy. This also explains section 8: with $m_1 = 0$ the cumulants have nowhere to live, and a cone supplies the missing m_1 .

Question 9.5.

- (i) Into the square-zero cone, whiskering is an A_∞ -functor with two components (proposition 9.4). Into the free cone, do the higher Boolean cumulants b_n , with generators of degree $1 - n$, give the higher components, as in the bar-construction description of Boolean cumulants?
- (ii) Classical and free cumulants are organized by set partitions and non-crossing partitions, and A_∞ -structures by associahedra. The permutohedron is a linear image of the Birkhoff polytope, and its faces correspond to ordered set partitions. Uniform square coupling polytopes are scaled Birkhoff polytopes, which lie on the walls of the chamber structure. Does cumulant combinatorics live on these degenerate walls?

10 The coupling nerve

The theory paper [2] organizes couplings into a simplicial object. This section computes its low-dimensional pieces. The *coupling nerve* on a finite set Y is

$$N_k = D(Y^{k+1}),$$

with face maps d_i that forget the i th coordinate (marginalization) and degeneracies s_i that repeat it (pushforward along the diagonal). It is D applied levelwise to the contractible Kan complex $Y^{\bullet+1}$. The relative version over a 1-cell, with kernels $X \rightarrow D(Y_0 \times \cdots \times Y_k)$, works pointwise in x . A 1-simplex is a coupling, and a 2-simplex with faces $\alpha = d_2$, $\beta = d_0$ is a joint law whose d_1 face is a composite of α and β . Every claim below is checked in `check_nerve.py`.

10.1 2-horns: fillers form toric polytopes

Proposition 10.1 (2-horn fibres). *Let p_{01} and p_{12} be compatible couplings on $Y_0 \times Y_1$ and $Y_1 \times Y_2$, with common marginal p_1 . The fillers, meaning laws p on $Y_0 \times Y_1 \times Y_2$ with these two faces, form the product over x_1 of the transportation polytopes of the slices $p(\cdot, x_1, \cdot)$, which have marginals $p_{01}(\cdot, x_1)$ and $p_{12}(x_1, \cdot)$. The dimension is $|Y_1|(|Y_0| - 1)(|Y_2| - 1)$, and the fibre is Delzant when every slice is nondegenerate: a product of the toric manifolds of section 2.*

Proof. Every marginal constraint involves a single value of x_1 , so the constraints decouple. The script checks the decoupling and the dimension for shapes $2 \times 2 \times 2$, $2 \times 3 \times 3$ and $3 \times 2 \times 3$, and Delzant is theorem 2.1 for each slice. $\square$

Proposition 10.2 (The canonical filler). *The gluing filler*

$$p^{\text{gl}}(x_0, x_1, x_2) = p_{01}(x_0, x_1) p_{12}(x_1, x_2) / p_1(x_1)$$

is the unique filler of maximal entropy. Its d_1 face is the vertical composite of the two couplings. Any filler differs from it slice by slice by $p_1(x_1) \text{Cov}(1_{\{x_0\}}, 1_{\{x_2\}} \mid x_1)$, a circulation.

Proof. $\log p^{\text{gl}}$ is a sum of a function of (x_0, x_1) and a function of (x_1, x_2) , so it is orthogonal to every tangent vector of the fibre. Hence p^{gl} is a critical point of the strictly concave entropy on the fibre, and so its unique maximizer. This is the inequality $H(X_0 X_1 X_2) \leq H(X_0 X_1) + H(X_1 X_2) - H(X_1)$, with equality exactly under conditional independence. The covariance identity is direct. The script checks orthogonality to a basis of the tangent space, a strict decrease of entropy along random feasible directions, and the identity in exact arithmetic. $\square$

So composition of 2-cells is the canonical (maximum-entropy) filling of 2-horns, and the other fillers are the same cumulant directions as in [section 9](#), now taken conditionally on the middle vertex. This is the 2-horn analogue of the Birch point of [proposition 9.1](#).

10.2 Higher horns do not always fill

Proposition 10.3 (Parity horns). *On bits, let a horn Λ_k^n carry, on each of its n faces, the uniform law on assignments of fixed parity, with exactly one face odd. These data agree on all overlaps. For $n = 2$ every such horn fills. For every $n = 3$ horn the contextual fraction of the horn, computed as the minimal unmodeled weight with global assignments as sources [[3](#), Thm. 7.12], is $\frac{1}{2}$. For every $n = 4$ horn it is $\frac{3}{4}$. Both values come with exactly verified certificates. The best approximate fillers are at distance $\varepsilon^* = \frac{1}{6}$ for $n = 3$ and $\varepsilon^* = \frac{3}{16}$ for $n = 4$.*

Corollary 10.4. *The coupling nerve $D(Y^{\bullet+1})$ is not a Kan complex, although $Y^{\bullet+1}$ is. All 2-horns fill, and the obstruction to filling higher horns is measured by the contextual fraction. This matches Vorob'ev's theorem [[22](#)]: every compatible family on a cover extends exactly when the cover is acyclic, which holds for the two faces of a 2-horn and fails for the faces of an n -horn with $n \geq 3$.*

Compatible 3-horns that are mixtures $t \cdot (\text{parity}) + (1 - t) \cdot (\text{restriction of a global law})$ have contextual fraction at most $t/2$. In all 40 random samples the inequality was strict.

10.3 Boundary fibres: where the toric picture stops

Filling the whole boundary $\partial\Delta^2$ means fixing all three 2-margins of a 3-way table. The fibre is a planar 3-way transportation polytope of dimension $(n_0 - 1)(n_1 - 1)(n_2 - 1)$.

Proposition 10.5 (Boundary fibres).

- (i) *For shape $2 \times n \times m$, the table is determined by its first layer $y = x(0, \cdot, \cdot)$, subject to fixed row and column sums and $0 \leq y \leq m_{12}$. So the fibre is a capacitated transportation polytope. Its constraint matrix is totally unimodular, so it is Delzant whenever it is simple. All generic samples of shape $2 \times 2 \times 2$, $2 \times 2 \times 3$ and $2 \times 3 \times 3$ were simple and smooth.*
- (ii) *For shape $3 \times 3 \times 3$ (exploratory), generic samples are simple but not smooth. In one instance with 1751 vertices, 202 vertices have edge vectors with entries ± 2 whose lattice has index 64 (Smith invariants $1, 1, 2, 2, 2, 2, 2$). The fibre is therefore the moment polytope of a toric orbifold [[17](#)], not of a manifold.*
- (iii) *By De Loera and Onn, every rational polytope is a planar $r \times s \times 3$ transportation polytope [[8](#)]. So boundary fibres eventually realize arbitrary polytopes, and no toric regularity can hold for them in general.*

Proof. (i) The (j, k) -margins give $x(1, j, k) = m_{jk} - y_{jk}$, and the other margins become the row and column sums of y . The constraint matrix is a bipartite incidence matrix stacked with an

identity, which is totally unimodular, so edges are ± 1 cycle vectors, as in [theorem 2.1](#). (ii) Vertices are enumerated exactly by simplex pivoting, adjacency is read off from zero patterns, and lattice indices are computed by Smith normal form. (iii) is the cited theorem. $\square$

Filling problem	Fibre	Toric status
1-simplex over its two vertices	coupling polytope	Delzant if nondegenerate (theorem 2.1)
2-horn Λ_k^2	product of coupling polytopes	Delzant if slices nondegenerate (proposition 10.1)
$\partial\Delta^2$, shape $2 \times n \times m$	capacitated transportation	Delzant if simple (proposition 10.5)
$\partial\Delta^2$, shape $3 \times 3 \times 3$	planar 3-way table	simple, not smooth: an orbifold
$\partial\Delta^2$, general	planar 3-way table	arbitrary rational polytopes [8]
Λ_k^n , $n \geq 3$	may be empty	obstruction = contextual fraction (proposition 10.3)

A Changes from the earlier draft

Earlier claim	Status
22 chambers; 4/12/6 quadrilaterals, pentagons, hexagons	18 chambers; 2 triangles, 9 quadrilaterals, 6 pentagons, 1 hexagon (theorem 3.1). The accompanying script printed 18; a coordinate-order bug in its <code>intersect</code> produced wrong vertex counts.
Example 4.2 a quadrilateral	a triangle, $\mathbb{CP}^2$ (example 3.4)
Self-intersection table	corrected; the earlier patterns violated $\sum D_i^2 = 12 - 3k$
Flips; type change across $f_1 = \frac{1}{2}$	neither occurs (proposition 3.3)
Costs $c(\sigma, e_0) = c(e_0, \tau) = \frac{1}{2}$	both are 1; cost is a maximum over x
Postcomposition running h independently on equal pairs	not unital; replaced by definition 4.2 . The earlier example then has zero defect, and new examples are nonzero (proposition 4.3)
Flattening surjective; Example 10.10 constant	not surjective; for point masses it is the identity (proposition 5.2)
Uniform level two monotone iff every cell is a facet; $n = 2$ not monotone	uniform $n \times n$ is degenerate; $n = 2$ is monotone, $n \geq 3$ is not simple
Level-two law restricted to couplings (Thm. 10.13)	ill-typed; replaced by proposition 5.2(v)
Facets are cycles of $K_{n,n}$; 15 facets; smooth Fano fourfold	facets are cells; 9 facets; uniform 3×3 is not simple; nondegenerate square $n \geq 3$ is not Fano (proposition 6.3)
Lattice distances are sums over cycles	they are the entries of the coupling at the fibre position
Critical points of W are vertices (committed states)	critical points are interior torus fibres with flat connections (proposition 6.1); vertices are fixed points and optimal-transport optima (proposition 6.4)
$F_2 = \text{Hess } W$; Picard–Lefschetz formula; “symbolic proof”	withdrawn (section 8); the cited verification compared 0.1152 with 2
Cycle counts $N_{2k} = \frac{1}{2k} \binom{n}{k}^2 k!(k-1)!$; $K_{4,4}$ total 156	$N_{2k} = \binom{n}{k}^2 k!(k-1)!/2$; $K_{4,4}$ has $36 + 96 + 72 = 204$; $K_{5,5}$ has $100 + 600 + 1800 + 1440$
Bridgeland 2008 for all surfaces; “(i)–(iii) none known”	Bridgeland 2008 is for K3; general surfaces are [1]; (i) is answered by proposition 7.1

B Computational checks

`check_nerve.py` runs 10 checks, all passing: A1–A2 for [propositions 10.1](#) and [10.2](#), A3–A4 for [proposition 10.3](#), A5–A6 for [proposition 10.5](#) (A6 takes about a minute), and D1–D3 for [proposition 9.4](#).

`check_cumulants.py` runs 8 checks for [section 9](#), all passing: Q1–Q3 for [propositions 9.1](#) and [9.2](#), and E1–E5 for [proposition 9.3](#).

`check_polytope_companion.py` runs 26 checks, all passing:

- chambers, types, surfaces, the sum rule, orbits, the del Pezzo fan, wall-crossings and [example 3.4](#) (F1–F7);
- [propositions 2.2](#) and [3.5](#) in all 18 chambers (K1–K3);
- [propositions 4.1](#) and [4.3](#) (S1–S4);
- [propositions 5.1](#) and [5.2](#) (L1–L7);
- [proposition 6.1](#), the non-monotone examples and [proposition 6.3](#) (W1–W3);
- [proposition 6.4](#) by linear programming (T1);
- the cycle counts of $K_{3,3}$ and $K_{4,4}$ by enumeration (C1).

Rational arithmetic is exact. The critical points are solved symbolically, and the linear programs use floating point.

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